

Hull Tactical Market Prediction

Kaggle Competition: [Hull Tactical Market Prediction](#)

This notebook implements a complete pipeline for predicting S&P 500 daily excess returns and converting them into optimal portfolio weights in $[0, 2]$. The objective is to maximise the competition's adjusted Sharpe ratio, which penalises both excess volatility and sustained underperformance relative to the market index.

Key results: Adjusted Sharpe **1.0915** on a strict temporal holdout (1 500 days, 200-day purge gap), representing a +53% improvement over the baseline (weight = 1.0) score of 0.7151.

Assessment data rule: the workflow uses only `data/train.csv` ; Kaggle `test.csv` is excluded because it has no target.

Workflow

1. Load libraries, plotting style, and fixed ex-ante constants.
2. Define the competition score and utility functions used across the notebook.
3. Load data and create the strict outer split: tuning block, purge gap, and final holdout.
4. Build engineered features with a train-only fit and leakage-safe transform.
5. Inspect engineered features with intermediate tables and diagnostic plots.
6. Fit the regime detector (Gaussian HMM, 3 states), train the return model (XGBoost with confidence intervals), and define the strategy sizing logic.
7. Build the inner walk-forward CV cache on the tuning block and define the CV evaluation function.
8. Run an Optuna strategy-only search on cached CV folds and inspect study results.
9. Refit the pipeline on the full tuning block, evaluate once on the untouched holdout, and build performance series.
10. Report selected strategy parameters, best inner CV score, final holdout score, and holdout plots.

```
In [21]: # 1. Load libraries, plotting style, and fixed ex-ante constants.
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import optuna
import pandas as pd
from IPython.display import display
from matplotlib.ticker import PercentFormatter
from sklearn.ensemble import GradientBoostingRegressor, RandomForestRegressor
from sklearn.mixture import GaussianMixture
from sklearn.preprocessing import StandardScaler

from xgboost import XGBRegressor
from hmmlearn.hmm import GaussianHMM
HAS_XGBOOST = True
HAS_HMMLEARN = True

plt.style.use("seaborn-v0_8-whitegrid")
plt.rcParams.update(
    {
        "figure.figsize": (11, 5),
        "axes.spines.top": False,
        "axes.spines.right": False,
        "axes.titlesize": 14,
        "axes.labelsize": 11,
        "font.size": 11,
        "legend.frameon": False,
    }
)
COLORS = {
    "blue": "#1f4e79",
    "teal": "#2a9d8f",
    "gold": "#e9c46a",
    "red": "#d62828",
    "gray": "#6c757d",
}
optuna.logging.set_verbosity(optuna.logging.WARNING)

TARGET_COLUMNS = ["forward_returns", "risk_free_rate", "market_forward_excess_returns"]
LAGGED_TARGET_COLUMNS = [
    "lagged_forward_returns",
```

```

        "lagged_risk_free_rate",
        "lagged_market_forward_excess_returns",
    ]
    VIX_COLUMNS = [f"V{i}" for i in range(1, 14)]
    REGIME_MULTIPLIERS = {
        "bull_low_vol": 1.30,
        "bear_high_vol": 0.50,
        "transition": 0.70,
        "normal": 1.00,
    }

    FEATURE_PARAMS = {
        "rolling_windows": (5, 20, 60, 252),
        "rolling_sharpe_windows": (20, 60),
        "rolling_autocorr_windows": (5, 20),
        "availability_threshold": 0.5,
        "history_buffer": 400,
        "min_periods": 5,
        "clip_feature_value": 10.0,
        "vix_rf_estimators": 300,
        "vix_random_state": 42,
    }

    # Fixed ex-ante modeling parameters for this notebook.
    MODEL_PARAMS = {
        "random_state": 42,
        "model_type": "xgboost",
        "n_estimators": 400,
        "max_depth": 4,
        "learning_rate": 0.05,
        "subsample": 0.8,
        "colsample_bytree": 0.8,
        "min_samples_leaf": 20,
    }

    FIXED_STRATEGY_DEFAULTS = {
        "min_weight": 0.0,
        "max_weight": 2.0,
        "vol_scale_min": 0.5,
        "vol_scale_max": 2.0,
        "vol_servo_window": 60,
        "hard_vol_cap": 1.20,
    }

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INITIAL_STRATEGY_PARAMS = {
    "confidence_threshold": 0.55,
    "gain": 80.0,
    "target_annual_vol": 0.16,
    "soft_vol_cap": 1.10,
    **FIXED_STRATEGY_DEFAULTS,
}
SEARCH_SPACE = {
    "confidence_threshold": (0.35, 0.75),
    "gain": (20.0, 180.0),
    "target_annual_vol": (0.08, 0.30),
    "soft_vol_cap": (1.00, 1.20),
}

TRAIN_PATH = Path("data/train.csv")
USE_EXTERNAL_VIX = False
OUTER_HOLDOUT_SIZE = 1500
OUTER_PURGE_GAP = 200
N_OPTUNA_TRIALS = 40

print(f"HAS_XGB00ST = {HAS_XGB00ST}")
print(f"HAS_HMMLEARN = {HAS_HMMLEARN}")
print(f"Optuna version = {optuna.__version__}")

```

```

HAS_XGB00ST = True
HAS_HMMLEARN = True
Optuna version = 4.8.0

```

In [22]: *# 2. Define the competition score and the small utility functions used across the notebook.*

```
class ParticipantVisibleError(Exception):
    pass

def score(solution: pd.DataFrame, submission: pd.DataFrame) -> float:
    if not pd.api.types.is_numeric_dtype(submission["prediction"]):
        raise ParticipantVisibleError("Predictions must be numeric")

    solution = solution.copy()
    solution["position"] = submission["prediction"].to_numpy()

    if solution["position"].max() > 2:
        raise ParticipantVisibleError(f"Position of {solution['position'].max()} exceeds maximum of 2")
    if solution["position"].min() < 0:
        raise ParticipantVisibleError(f"Position of {solution['position'].min()} below minimum of 0")

    solution["strategy_returns"] = (
        solution["risk_free_rate"] * (1 - solution["position"]) + solution["position"] * solution["forward_returns"]
    )

    strategy_excess_returns = solution["strategy_returns"] - solution["risk_free_rate"]
    strategy_excess_cumulative = (1 + strategy_excess_returns).prod()
    strategy_mean_excess_return = strategy_excess_cumulative ** (1 / len(solution)) - 1
    strategy_std = solution["strategy_returns"].std()
    if strategy_std == 0:
        raise ParticipantVisibleError("Division by zero, strategy std is zero")

    sharpe = strategy_mean_excess_return / strategy_std * np.sqrt(252)
    strategy_volatility = float(strategy_std * np.sqrt(252) * 100)

    market_excess_returns = solution["forward_returns"] - solution["risk_free_rate"]
    market_excess_cumulative = (1 + market_excess_returns).prod()
    market_mean_excess_return = market_excess_cumulative ** (1 / len(solution)) - 1
    market_std = solution["forward_returns"].std()
    market_volatility = float(market_std * np.sqrt(252) * 100)
    if market_volatility == 0:
        raise ParticipantVisibleError("Division by zero, market std is zero")

    excess_vol = max(0, strategy_volatility / market_volatility - 1.2)
    vol_penalty = 1 + excess_vol
```

```

return_gap_penalty = max(0, (market_mean_excess_return - strategy_mean_excess_return) * 100 * 252)
return_penalty = 1 + (return_gap_penalty ** 2) / 100

adjusted_sharpe = sharpe / (vol_penalty * return_penalty)
return min(float(adjusted_sharpe), 1_000_000)

def key_num(name: str) -> tuple[str, int]:
    digits = "".join(c for c in name if c.isdigit())
    return name[0], int(digits) if digits else 0

def autocorr(values: np.ndarray) -> float:
    x = np.asarray(values, dtype=float)
    x = x[np.isfinite(x)]
    if x.size < 3:
        return 0.0
    a, b = x[:-1], x[1:]
    if np.std(a) < 1e-12 or np.std(b) < 1e-12:
        return 0.0
    return float(np.corrcoef(a, b)[0, 1])

def walk_forward_splits(n_samples: int, n_splits: int, initial_train_size: int, validation_size: int, purge_gap: int):
    for i in range(n_splits):
        train_end = initial_train_size + i * validation_size
        val_start = train_end + purge_gap
        val_end = val_start + validation_size
        if val_end > n_samples:
            break
        yield np.arange(0, train_end), np.arange(val_start, val_end)

from xgboost import XGBRegressor
from hmmlearn.hmm import GaussianHMM
HAS_XGBOOST = True
HAS_HMMLEARN = True

```

```

In [23]: # 3. Load data/train.csv and create the strict outer split.
train_df = pd.read_csv(TRAIN_PATH)

n_rows = len(train_df)
if OUTER_HOLDOUT_SIZE + OUTER_PURGE_GAP >= n_rows:
    raise ValueError("OUTER_HOLDOUT_SIZE + OUTER_PURGE_GAP is too large for dataset length.")

split_idx = n_rows - OUTER_HOLDOUT_SIZE - OUTER_PURGE_GAP
tune_df = train_df.iloc[:split_idx].copy()
holdout_df = train_df.iloc[split_idx + OUTER_PURGE_GAP :].copy()

cv_params = {
    "n_splits": 6,
    "initial_train_size": max(1500, int(len(tune_df) * 0.5)),
    "validation_size": max(150, int(len(tune_df) * 0.08)),
    "purge_gap": min(150, max(60, int(len(tune_df) * 0.02))),
}

split_summary = pd.Series(
    {
        "total_rows": int(n_rows),
        "tuning_rows": int(len(tune_df)),
        "outer_purge_gap_rows": int(OUTER_PURGE_GAP),
        "holdout_rows": int(len(holdout_df)),
        "tuning_end_date_id": int(tune_df["date_id"].iloc[-1]),
        "holdout_start_date_id": int(holdout_df["date_id"].iloc[0]),
    }
)

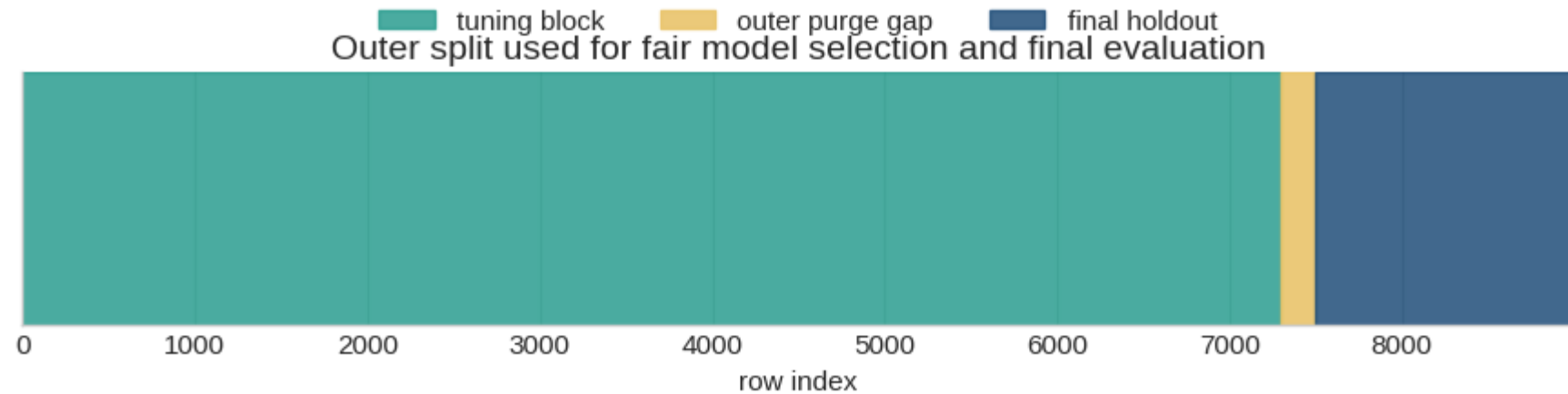
print("Split summary")
display(split_summary)
print("Inner CV parameters")
display(pd.Series(cv_params))

fig, ax = plt.subplots(figsize=(11, 1.8))
ax.axvspan(0, len(tune_df), color=COLORS["teal"], alpha=0.85, label="tuning block")
ax.axvspan(len(tune_df), len(tune_df) + OUTER_PURGE_GAP, color=COLORS["gold"], alpha=0.9, label="outer purge gap")
ax.axvspan(len(tune_df) + OUTER_PURGE_GAP, n_rows, color=COLORS["blue"], alpha=0.85, label="final holdout")
ax.set_xlim(0, n_rows)
ax.set_yticks([])
ax.set_xlabel("row index")
ax.set_title("Outer split used for fair model selection and final evaluation")

```

```
ax.legend(loc="upper center", ncol=3, bbox_to_anchor=(0.5, 1.35))  
plt.show()
```

Split summary
total_rows 8990
tuning_rows 7290
outer_purge_gap_rows 200
holdout_rows 1500
tuning_end_date_id 7289
holdout_start_date_id 7490
dtype: int64
Inner CV parameters
n_splits 6
initial_train_size 3645
validation_size 583
purge_gap 145
dtype: int64



In [24]: *# 4. Build the engineered features with a train-only fit and a leakage-safe transform.*

```
def ensure_lagged_targets(df: pd.DataFrame) -> pd.DataFrame:
    df = df.copy()
    mapping = {
        "lagged_forward_returns": "forward_returns",
        "lagged_risk_free_rate": "risk_free_rate",
        "lagged_market_forward_excess_returns": "market_forward_excess_returns",
    }
    for lag_col, source_col in mapping.items():
        if lag_col not in df.columns and source_col in df.columns:
            df[lag_col] = df[source_col].shift(1)
    return df

def fit_feature_state(train_df: pd.DataFrame) -> dict:
    raw_feature_columns = [
        c for c in train_df.columns if c not in (TARGET_COLUMNS + ["date_id", "is_scored"] + LAGGED_TARGET_COLUMNS)
    ]

    vix_model = None

    state = {
        "raw_feature_columns": raw_feature_columns,
        "vix_model": vix_model,
        "feature_columns": None,
        "medians": None,
        "sorted_vix_proxy": None,
        "return_history": None,
        "vix_history": None,
        "macro_mean": 0.0,
        "macro_std": 1.0,
        "sentiment_mean": 0.0,
        "sentiment_std": 1.0,
        "price_mean": 0.0,
        "price_std": 1.0,
    }

    provisional = build_feature_frame(train_df, state, fit_mode=True, use_history=False)
    provisional = provisional.ffill().replace([np.inf, -np.inf], np.nan)
    medians = provisional.median(numeric_only=True)
    provisional = provisional.fillna(medians).fillna(0.0)
```

```

state["medians"] = medians
state["feature_columns"] = provisional.columns.tolist()
state["return_history"] = provisional.get("lagged_forward_returns", pd.Series(dtype=float)).tail(FEATURE_PARAMS["history_buffer"])
state["vix_history"] = provisional.get("vix_proxy", pd.Series(dtype=float)).tail(FEATURE_PARAMS["history_buffer"])
return state

```

```

def build_feature_frame(df: pd.DataFrame, state: dict, fit_mode: bool, use_history: bool) -> pd.DataFrame:
    out = ensure_lagged_targets(df)

```

```

    raw_feature_columns = state["raw_feature_columns"]
    availability = out[raw_feature_columns].notna().mean(axis=1)
    out["data_availability"] = (availability > FEATURE_PARAMS["availability_threshold"]).astype(float)

```

```

    vix_proxy = out[VIX_COLUMNS].mean(axis=1)
    out["vix_proxy"] = vix_proxy

```

```

    out["vix_term_slope_proxy"] = out["V13"] - out["V1"]

```

```

    if fit_mode:
        out["vix_proxy_pct"] = out["vix_proxy"].rank(pct=True)
        state["sorted_vix_proxy"] = np.sort(out["vix_proxy"].dropna().to_numpy())
        out["vix_proxy_delta"] = out["vix_proxy"].diff()

```

```

    else:
        sorted_ref = state["sorted_vix_proxy"]
        pos = np.searchsorted(sorted_ref, out["vix_proxy"].to_numpy(), side="right")
        out["vix_proxy_pct"] = pos / float(len(sorted_ref))
        if use_history:
            joined = pd.concat([state["vix_history"], out["vix_proxy"]], ignore_index=True)
            out["vix_proxy_delta"] = joined.diff().iloc[-len(out) :].to_numpy()
        else:
            out["vix_proxy_delta"] = out["vix_proxy"].diff()

```

```

    returns = out["lagged_forward_returns"].astype(float)
    if use_history:
        returns_full = pd.concat([state["return_history"], returns], ignore_index=True)

```

```

    else:
        returns_full = returns.reset_index(drop=True)

```

```

    n = len(returns)
    for window in FEATURE_PARAMS["rolling_windows"]:
        vol = returns_full.rolling(window=window, min_periods=FEATURE_PARAMS["min_periods"]).std(ddof=0) * np.sqrt(252)
        mom = (1.0 + returns_full.fillna(0.0)).rolling(window=window, min_periods=FEATURE_PARAMS["min_periods"]).apply(lambda x: x.rolling(window=window, min_periods=1).sum().iloc[-1])

```

```

out[f"roll_vol_{window}"] = vol.iloc[-n:].to_numpy()
out[f"momentum_{window}"] = mom.iloc[-n:].to_numpy()

for window in FEATURE_PARAMS["rolling_sharpe_windows"]:
    mean_r = returns_full.rolling(window=window, min_periods=FEATURE_PARAMS["min_periods"]).mean()
    std_r = returns_full.rolling(window=window, min_periods=FEATURE_PARAMS["min_periods"]).std(ddof=0)
    out[f"roll_sharpe_{window}"] = (mean_r / (std_r + 1e-8) * np.sqrt(252)).iloc[-n:].to_numpy()

cumulative = (1.0 + returns_full.fillna(0.0)).cumprod()
rolling_high_60 = cumulative.rolling(window=60, min_periods=1).max()
drawdown = cumulative / rolling_high_60 - 1.0
out["drawdown_60"] = drawdown.iloc[-n:].to_numpy()
out["max_drawdown_252"] = drawdown.rolling(window=252, min_periods=1).min().iloc[-n:].to_numpy()
rolling_vol_20 = returns_full.rolling(window=20, min_periods=FEATURE_PARAMS["min_periods"]).std(ddof=0)
out["vol_of_vol_20"] = rolling_vol_20.rolling(window=20, min_periods=FEATURE_PARAMS["min_periods"]).std(ddof=0).iloc[-n:].to_numpy()

for window in FEATURE_PARAMS["rolling_autocorr_windows"]:
    ac = returns_full.rolling(window=window, min_periods=FEATURE_PARAMS["min_periods"]).apply(autocorr, raw=True)
    out[f"autocorr_{window}"] = ac.iloc[-n:].to_numpy()

i_cols = sorted([c for c in out.columns if c.startswith("I")], key=key_num)
e_cols = [c for c in out.columns if c.startswith("E")]
s_cols = [c for c in out.columns if c.startswith("S")]
p_cols = [c for c in out.columns if c.startswith("P")]

out["yield_curve_proxy"] = out[i_cols[-1]] - out[i_cols[0]]
macro = out[e_cols].mean(axis=1)
sentiment = out[s_cols].mean(axis=1)
price = out[p_cols].mean(axis=1)

if fit_mode:
    state["macro_mean"] = float(macro.mean())
    state["macro_std"] = float(macro.std(ddof=0) + 1e-8)
    state["sentiment_mean"] = float(sentiment.mean())
    state["sentiment_std"] = float(sentiment.std(ddof=0) + 1e-8)
    state["price_mean"] = float(price.mean())
    state["price_std"] = float(price.std(ddof=0) + 1e-8)

macro_z = (macro - state["macro_mean"]) / state["macro_std"]
sentiment_z = (sentiment - state["sentiment_mean"]) / state["sentiment_std"]
price_z = (price - state["price_mean"]) / state["price_std"]

out["macro_sentiment_divergence"] = macro_z - sentiment_z

```

```

momentum_20 = out.get("momentum_20", pd.Series(0.0, index=out.index))
out["price_momentum_agreement"] = np.sign(price_z * momentum_20.fillna(0.0)).replace(0, 1.0).astype(float)

out = out.drop(columns=[c for c in VIX_COLUMNS if c in out.columns], errors="ignore")
out = out.drop(columns=["date_id", "is_scored"], errors="ignore")
out = out.drop(columns=[c for c in TARGET_COLUMNS if c in out.columns], errors="ignore")
return out

```

```

def transform_features(df: pd.DataFrame, state: dict, use_history: bool = True) -> pd.DataFrame:
    features = build_feature_frame(df, state, fit_mode=False, use_history=use_history)
    features = features.ffill()
    features = features.fillna(state["medians"])
    features = features.replace([np.inf, -np.inf], np.nan).fillna(0.0)
    features = features.clip(-FEATURE_PARAMS["clip_feature_value"], FEATURE_PARAMS["clip_feature_value"])
    return features[state["feature_columns"]]

```

```

In [25]: # 5. Inspect the engineered features with intermediate tables and diagnostic plots.
feature_preview_state = fit_feature_state(tune_df)
X_tune_preview = transform_features(tune_df, feature_preview_state, use_history=False)
X_holdout_preview = transform_features(holdout_df, feature_preview_state, use_history=True)

feature_summary = pd.Series(
    {
        "engineered_feature_count": int(X_tune_preview.shape[1]),
        "tuning_rows_after_feature_step": int(X_tune_preview.shape[0]),
        "holdout_rows_after_feature_step": int(X_holdout_preview.shape[0]),
        "missing_share_tuning": float(X_tune_preview.isna().mean().mean()),
        "mean_roll_vol_20": float(X_tune_preview.get("roll_vol_20", pd.Series(dtype=float)).mean()),
        "mean_momentum_20": float(X_tune_preview.get("momentum_20", pd.Series(dtype=float)).mean()),
    }
)

print("Feature summary")
display(feature_summary)
print("Preview of engineered features")
display(
    X_tune_preview[
        [c for c in ["lagged_forward_returns", "vix_proxy", "roll_vol_20", "momentum_20", "drawdown_60", "macro_sentimen
        ].head(10)
    )

recent_feature_view = X_tune_preview[
    [c for c in ["vix_proxy", "roll_vol_20", "momentum_20", "drawdown_60"] if c in X_tune_preview.columns]
].tail(300).copy()
fig, axes = plt.subplots(2, 1, figsize=(11, 7), sharex=True)
recent_feature_view[[c for c in ["vix_proxy", "roll_vol_20"] if c in recent_feature_view.columns]].plot(
    ax=axes[0],
    color=[COLORS["red"], COLORS["blue"]][: len([c for c in ["vix_proxy", "roll_vol_20"] if c in recent_feature_view.co
    linewidth=2,
)
axes[0].set_title("Recent engineered risk features on the tuning block")
axes[0].set_ylabel("level")
recent_feature_view[[c for c in ["momentum_20", "drawdown_60"] if c in recent_feature_view.columns]].plot(
    ax=axes[1],
    color=[COLORS["teal"], COLORS["gray"]][: len([c for c in ["momentum_20", "drawdown_60"] if c in recent_feature_view
    linewidth=2,
)
axes[1].set_title("Recent engineered trend features on the tuning block")

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```

axes[1].set_ylabel("level")
axes[1].set_xlabel("row index within tuning block")
plt.tight_layout()
plt.show()

```

Feature summary

```

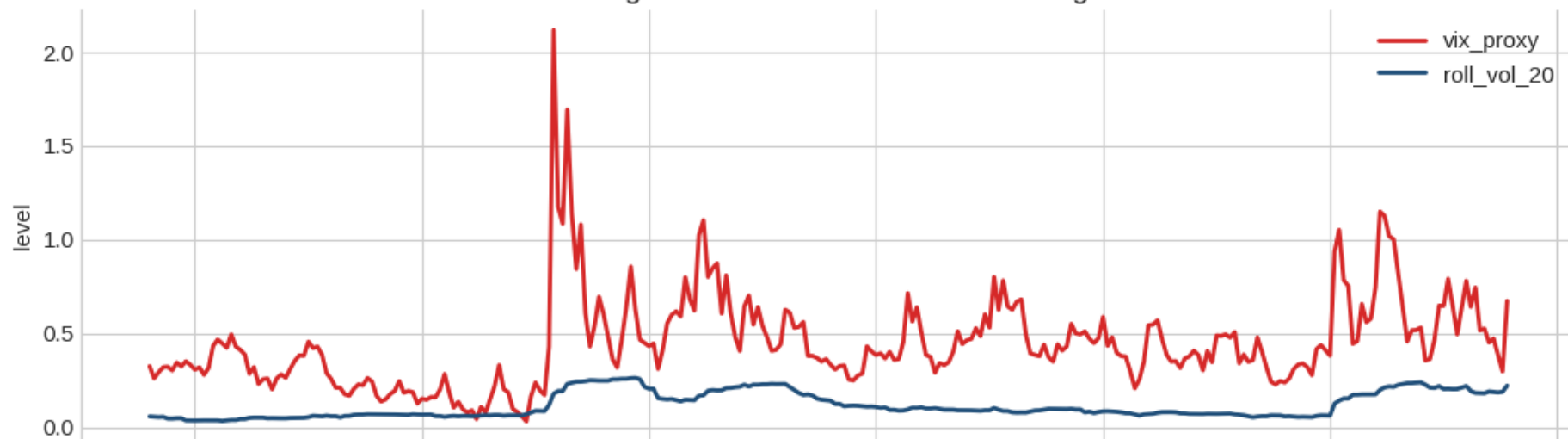
engineered_feature_count      107.000000
tuning_rows_after_feature_step 7290.000000
holdout_rows_after_feature_step 1500.000000
missing_share_tuning          0.000000
mean_roll_vol_20              0.144780
mean_momentum_20              0.008286
dtype: float64

```

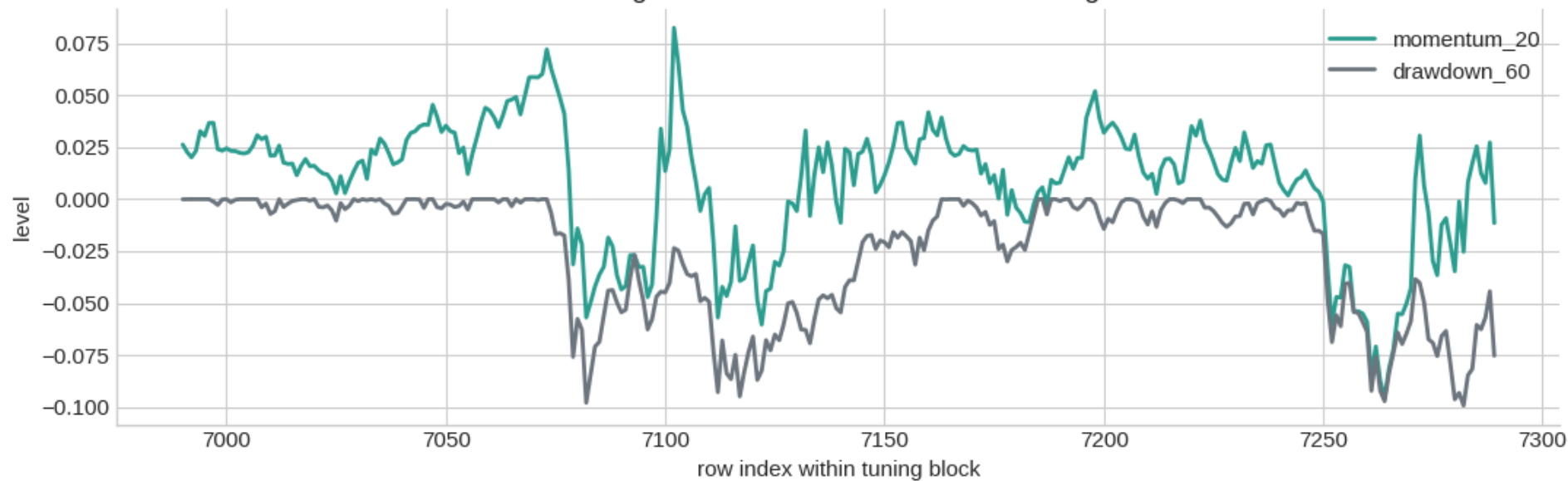
Preview of engineered features

	lagged_forward_returns	vix_proxy	roll_vol_20	momentum_20	drawdown_60	macro_sentiment_divergence
0	0.000631	0.245208	0.127185	0.012145	0.000000	0.051732
1	-0.002421	0.245208	0.127185	0.012145	-0.002421	0.051732
2	-0.008495	0.245208	0.127185	0.012145	-0.010895	0.051732
3	-0.009624	0.245208	0.127185	0.012145	-0.020415	0.051732
4	0.004662	0.245208	0.127185	-0.015848	-0.015848	0.051732
5	-0.011686	0.245208	0.094449	-0.027348	-0.027348	0.051732
6	-0.006449	0.245208	0.086398	-0.033621	-0.033621	0.051732
7	0.003644	0.245208	0.095260	-0.030100	-0.030100	0.051732
8	-0.024566	0.245208	0.138627	-0.053926	-0.053926	0.051732
9	-0.008492	0.245208	0.130950	-0.061960	-0.061960	0.051732

Recent engineered risk features on the tuning block



Recent engineered trend features on the tuning block



In [26]: *# 6. Fit the regime detector, fit the return model, and define the strategy sizing logic.*

```
def fit_regime_detector(X: pd.DataFrame, returns: pd.Series | None = None, n_states: int = 3) -> dict:
    feature_columns = ["vix_proxy_pct", "roll_vol_20", "momentum_20", "autocorr_20"]

    regime_input = X[feature_columns].copy().replace([np.inf, -np.inf], np.nan).fillna(0.0)
    scaler = StandardScaler()
    Z = scaler.fit_transform(regime_input)

    resolved_states = int(max(2, min(n_states, len(regime_input) // 200 if len(regime_input) > 400 else 2)))
    model = GaussianHMM(n_components=resolved_states, covariance_type="diag", n_iter=250, random_state=42)
    model.fit(Z)
    states = model.predict(Z)
    probs = model.predict_proba(Z)
    transition_matrix = model.transmat_

    if returns is None:
        state_labels = {s: f"state_{s}" for s in range(int(states.max()) + 1)}
    else:
        stats = (
            pd.DataFrame({"state": states, "ret": pd.Series(returns).reset_index(drop=True).astype(float)})
            .groupby("state")["ret"]
            .agg(["mean", "std", "count"])
            .rename(columns={"std": "vol"})
        )
        stats["vol"] = stats["vol"].fillna(0.0)
        best_ret_state = int(stats["mean"].idxmax())
        worst_ret_state = int(stats["mean"].idxmin())
        highest_vol_state = int(stats["vol"].idxmax())
        state_labels = {s: "normal" for s in range(int(states.max()) + 1)}
        if best_ret_state != highest_vol_state:
            state_labels[best_ret_state] = "bull_low_vol"
            state_labels[worst_ret_state] = "bear_high_vol"
        for s in state_labels:
            if s not in (best_ret_state, worst_ret_state):
                state_labels[s] = "transition"

    return {
        "feature_columns": feature_columns,
        "scaler": scaler,
        "model": model,
        "state_labels": state_labels,
        "transition_matrix": transition_matrix,
```



```
}
```

```
def predict_regimes(X: pd.DataFrame, state: dict) -> pd.DataFrame:
    Z = X[state["feature_columns"].copy().replace([np.inf, -np.inf], np.nan).fillna(0.0)]
    Z = state["scaler"].transform(Z)

    probs = state["model"].predict_proba(Z)
    states = state["model"].predict(Z)

    out = pd.DataFrame(index=X.index)
    out["regime_state"] = states
    for i in range(probs.shape[1]):
        out[f"regime_prob_{i}"] = probs[:, i]
    out["regime_stability"] = probs.max(axis=1)
    out["regime_switch_risk"] = 1.0 - np.array([state["transition_matrix"][s, s] for s in states], dtype=float)
    out["regime_label"] = [state["state_labels"].get(int(s), f"state_{int(s)}") for s in states]
    return out

def fit_return_model(X: pd.DataFrame, y: pd.Series) -> dict:
    X = X.copy().replace([np.inf, -np.inf], np.nan).fillna(0.0)
    y = pd.Series(y).astype(float)

    point_model = XGBRegressor(
        n_estimators=MODEL_PARAMS["n_estimators"],
        max_depth=MODEL_PARAMS["max_depth"],
        learning_rate=MODEL_PARAMS["learning_rate"],
        subsample=MODEL_PARAMS["subsample"],
        colsample_bytree=MODEL_PARAMS["colsample_bytree"],
        objective="reg:squarederror",
        tree_method="hist",
        seed=MODEL_PARAMS["random_state"],
        nthread=1,
    )
    lower_model = XGBRegressor(
        n_estimators=max(150, MODEL_PARAMS["n_estimators"] // 3),
        max_depth=MODEL_PARAMS["max_depth"],
        learning_rate=MODEL_PARAMS["learning_rate"],
        subsample=MODEL_PARAMS["subsample"],
        colsample_bytree=MODEL_PARAMS["colsample_bytree"],
        objective="reg:quantileerror",
        quantile_alpha=0.1,
```

```

        tree_method="hist",
        seed=MODEL_PARAMS["random_state"],
        nthread=1,
    )
    upper_model = XGBRegressor(
        n_estimators=max(150, MODEL_PARAMS["n_estimators"] // 3),
        max_depth=MODEL_PARAMS["max_depth"],
        learning_rate=MODEL_PARAMS["learning_rate"],
        subsample=MODEL_PARAMS["subsample"],
        colsample_bytree=MODEL_PARAMS["colsample_bytree"],
        objective="reg:quantileerror",
        quantile_alpha=0.9,
        tree_method="hist",
        seed=MODEL_PARAMS["random_state"],
        nthread=1,
    )

```

```

point_model.fit(X, y)
lower_model.fit(X, y)
upper_model.fit(X, y)

```

```

train_pred = point_model.predict(X)
train_low = lower_model.predict(X)
train_high = upper_model.predict(X)
width = np.maximum(train_high - train_low, 1e-8)
raw_conf = np.abs(train_pred) / width

```

```

return {
    "feature_columns": X.columns.tolist(),
    "point_model": point_model,
    "lower_model": lower_model,
    "upper_model": upper_model,
    "confidence_low": float(np.quantile(raw_conf, 0.1)),
    "confidence_high": float(np.quantile(raw_conf, 0.9)),
}

```

```

def predict_return_model(X: pd.DataFrame, state: dict) -> pd.DataFrame:
    X = X.copy()
    X = X[state["feature_columns"]].replace([np.inf, -np.inf], np.nan).fillna(0.0)

    point = state["point_model"].predict(X)
    lower = state["lower_model"].predict(X)

```

```

upper = state["upper_model"].predict(X)
width = np.maximum(upper - lower, 1e-8)
raw_conf = np.abs(point) / width
confidence = (raw_conf - state["confidence_low"]) / (state["confidence_high"] - state["confidence_low"] + 1e-8)
confidence = np.clip(confidence, 0.0, 1.0)

return pd.DataFrame(
    {
        "predicted_excess_return": point,
        "confidence": confidence,
    },
    index=X.index,
)

```

```

def volatility_ratio(strategy_returns: pd.Series, market_returns: pd.Series) -> float:
    std_s = float(strategy_returns.std(ddof=0))
    std_m = float(market_returns.std(ddof=0))
    return std_s / (std_m + 1e-8)

```

```

def return_gap(strategy_returns: list[float], market_returns: list[float]) -> float:
    s = np.array(strategy_returns, dtype=float)
    m = np.array(market_returns, dtype=float)
    strat_mean = np.prod(1.0 + s) ** (1.0 / len(s)) - 1.0
    mkt_mean = np.prod(1.0 + m) ** (1.0 / len(m)) - 1.0
    return max(0.0, (mkt_mean - strat_mean) * 252)

```

```

def generate_weights(
    predicted_excess_return: pd.Series,
    confidence: pd.Series,
    strategy_params: dict,
    current_volatility: pd.Series,
    regime_label: pd.Series,
    forward_returns: pd.Series,
    risk_free_rate: pd.Series,
) -> pd.Series:
    pred = pd.Series(predicted_excess_return).astype(float).reset_index(drop=True)
    conf = pd.Series(confidence).astype(float).reset_index(drop=True)
    vol = pd.Series(current_volatility).astype(float).reset_index(drop=True)
    regimes = pd.Series(regime_label).astype(str).reset_index(drop=True)

```

```

params = {**FIXED_STRATEGY_DEFAULTS, **strategy_params}
weights = np.ones(len(pred), dtype=float)
strat_returns = []
mkt_returns = []

for i in range(len(pred)):
    if conf.iloc[i] < params["confidence_threshold"]:
        weight = 1.0
    else:
        reg_mult = REGIME_MULTIPLIERS.get(regimes.iloc[i], 1.0)
        annual_vol = float(vol.iloc[i]) if np.isfinite(vol.iloc[i]) else params["target_annual_vol"]
        annual_vol = max(annual_vol, 1e-6)
        vol_scale = np.clip(
            params["target_annual_vol"] / annual_vol,
            params["vol_scale_min"],
            params["vol_scale_max"],
        )
        weight = 1.0 + params["gain"] * float(pred.iloc[i]) * vol_scale * reg_mult

    if i > 0:
        strat_hist = pd.Series(strat_returns[-params["vol_servo_window"] :], dtype=float)
        mkt_hist = pd.Series(mkt_returns[-params["vol_servo_window"] :], dtype=float)
        ratio = volatility_ratio(strat_hist, mkt_hist)
        if ratio > params["hard_vol_cap"]:
            weight = 1.0 + 0.20 * (weight - 1.0)
        elif ratio > params["soft_vol_cap"]:
            weight = 1.0 + 0.55 * (weight - 1.0)
        if return_gap(strat_returns, mkt_returns) > 0:
            weight = max(weight, 1.0)

    weight = float(np.clip(weight, params["min_weight"], params["max_weight"]))
    weights[i] = weight

    fwd = float(pd.Series(forward_returns).iloc[i])
    rf = float(pd.Series(risk_free_rate).iloc[i])
    excess_return = fwd - rf
    strat_returns.append(weight * excess_return + rf)
    mkt_returns.append(excess_return + rf)

return pd.Series(weights, index=predicted_excess_return.index)

```

```

In [27]: # 7. Build the inner walk-forward CV cache on the tuning block and define the CV evaluation function.
cv_fold_cache = []

for fold_number, (train_idx, val_idx) in enumerate(walk_forward_splits(len(tune_df), **cv_params), start=1):
    train_fold = tune_df.iloc[train_idx].copy()
    val_fold = tune_df.iloc[val_idx].copy()

    # 7.1 Fit the full fixed modeling pipeline on this fold once.
    feature_state = fit_feature_state(train_fold)
    X_train = transform_features(train_fold, feature_state, use_history=False)
    X_val = transform_features(val_fold, feature_state, use_history=True)

    regime_state = fit_regime_detector(X_train, returns=train_fold["forward_returns"], n_states=3)
    reg_train = predict_regimes(X_train, regime_state)
    reg_val = predict_regimes(X_val, regime_state)

    X_train_all = pd.concat([X_train, reg_train.drop(columns=["regime_label"])], axis=1)
    X_val_all = pd.concat([X_val, reg_val.drop(columns=["regime_label"])], axis=1)

    model_state = fit_return_model(X_train_all, train_fold["market_forward_excess_returns"])
    pred_val = predict_return_model(X_val_all, model_state)

    cv_fold_cache.append(
        {
            "fold": fold_number,
            "train_start": int(train_idx.min()),
            "train_end": int(train_idx.max()),
            "val_start": int(val_idx.min()),
            "val_end": int(val_idx.max()),
            "solution": val_fold[["risk_free_rate", "forward_returns"]].copy(),
            "predicted_excess_return": pred_val["predicted_excess_return"].copy(),
            "confidence": pred_val["confidence"].copy(),
            "current_volatility": X_val.get("roll_vol_20", pd.Series(np.nan, index=X_val.index)).copy(),
            "regime_label": reg_val["regime_label"].copy(),
        }
    )

def evaluate_strategy_params(strategy_params: dict, fold_cache: list[dict]) -> tuple[float, pd.DataFrame]:
    fold_rows = []
    fold_scores = []
    for fold_payload in fold_cache:

```

```

weights = generate_weights(
    predicted_excess_return=fold_payload["predicted_excess_return"],
    confidence=fold_payload["confidence"],
    strategy_params=strategy_params,
    current_volatility=fold_payload["current_volatility"],
    regime_label=fold_payload["regime_label"],
    forward_returns=fold_payload["solution"]["forward_returns"],
    risk_free_rate=fold_payload["solution"]["risk_free_rate"],
)
fold_score = score(
    fold_payload["solution"],
    pd.DataFrame({"prediction": weights.values}, index=fold_payload["solution"].index),
)
fold_scores.append(float(fold_score))
fold_rows.append(
    {
        "fold": fold_payload["fold"],
        "train_start": fold_payload["train_start"],
        "train_end": fold_payload["train_end"],
        "val_start": fold_payload["val_start"],
        "val_end": fold_payload["val_end"],
        "adjusted_sharpe": float(fold_score),
        "mean_confidence": float(fold_payload["confidence"].mean()),
        "mean_weight": float(weights.mean()),
        "mean_roll_vol_20": float(fold_payload["current_volatility"].mean()),
        "dominant_regime": fold_payload["regime_label"].mode().iloc[0],
    }
)
return float(np.mean(fold_scores)), pd.DataFrame(fold_rows)

baseline_inner_cv_mean, baseline_cv_results = evaluate_strategy_params(INITIAL_STRATEGY_PARAMS, cv_fold_cache)
print(f"Cached CV folds built: {len(cv_fold_cache)}")
print(f"Initial untuned strategy mean CV adjusted Sharpe: {baseline_inner_cv_mean:.4f}")
display(baseline_cv_results)

```

Cached CV folds built: 6

Initial untuned strategy mean CV adjusted Sharpe: 0.7246

	fold	train_start	train_end	val_start	val_end	adjusted_sharpe	mean_confidence	mean_weight	mean_roll_vol_20	dominant_regime
0	1	0	3644	3790	4372	0.976980	0.350858	0.964844	0.097941	transition
1	2	0	4227	4373	4955	-0.435979	0.423197	1.076260	0.237727	transition
2	3	0	4810	4956	5538	0.641542	0.369652	1.012409	0.177386	bear_high_vol
3	4	0	5393	5539	6121	1.610602	0.322459	1.003750	0.113536	transition
4	5	0	5976	6122	6704	0.513374	0.533008	1.008364	0.126502	bear_high_vol
5	6	0	6559	6705	7287	1.040961	0.887947	0.919900	0.097888	transition

```

In [28]: # 8. Run a narrow Optuna strategy-only search on the cached CV folds and inspect the study results.
def objective(trial: optuna.trial.Trial) -> float:
    strategy_params = {
        "confidence_threshold": trial.suggest_float("confidence_threshold", *SEARCH_SPACE["confidence_threshold"]),
        "gain": trial.suggest_float("gain", *SEARCH_SPACE["gain"]),
        "target_annual_vol": trial.suggest_float("target_annual_vol", *SEARCH_SPACE["target_annual_vol"]),
        "soft_vol_cap": trial.suggest_float("soft_vol_cap", *SEARCH_SPACE["soft_vol_cap"]),
        **FIXED_STRATEGY_DEFAULTS,
    }
    mean_score, _ = evaluate_strategy_params(strategy_params, cv_fold_cache)
    return mean_score

sampler = optuna.samplers.TPESampler(seed=42)
study = optuna.create_study(direction="maximize", sampler=sampler)
study.optimize(objective, n_trials=N_OPTUNA_TRIALS, show_progress_bar=False)

best_strategy_params = {**study.best_trial.params, **FIXED_STRATEGY_DEFAULTS}
best_inner_cv_mean_adjusted_sharpe, best_cv_results = evaluate_strategy_params(best_strategy_params, cv_fold_cache)

study_results = study.trials_dataframe(attrs=("number", "value", "params", "state")).rename(columns={"value": "mean_cv_adjusted_sharpe"})
study_results["best_so_far"] = study_results["mean_cv_adjusted_sharpe"].cummax()

print("Best Optuna trial parameters")
display(pd.Series(best_strategy_params))
print(f"Best inner CV mean adjusted Sharpe: {best_inner_cv_mean_adjusted_sharpe:.4f}")
print("Top trials")
display(study_results.sort_values("mean_cv_adjusted_sharpe", ascending=False).head(10))

fig, axes = plt.subplots(1, 2, figsize=(12.5, 4.5))
axes[0].plot(study_results["number"], study_results["mean_cv_adjusted_sharpe"], marker="o", linewidth=1.8, color=COLORS["teal"])
axes[0].plot(study_results["number"], study_results["best_so_far"], linewidth=2.4, color=COLORS["teal"], label="best so far")
axes[0].set_title("Optuna optimization history")
axes[0].set_xlabel("trial")
axes[0].set_ylabel("mean CV adjusted Sharpe")
axes[0].legend()
axes[1].bar(best_cv_results["fold"], best_cv_results["mean_weight"], color=COLORS["teal"], alpha=0.9, label="mean weight")
axes[1].plot(best_cv_results["fold"], best_cv_results["adjusted_sharpe"], marker="o", linewidth=2.2, color=COLORS["gold"])
axes[1].set_title("Best trial fold behaviour")
axes[1].set_xlabel("fold")
axes[1].set_ylabel("level")
axes[1].legend()

```



```
plt.tight_layout()
plt.show()
```

Best Optuna trial parameters

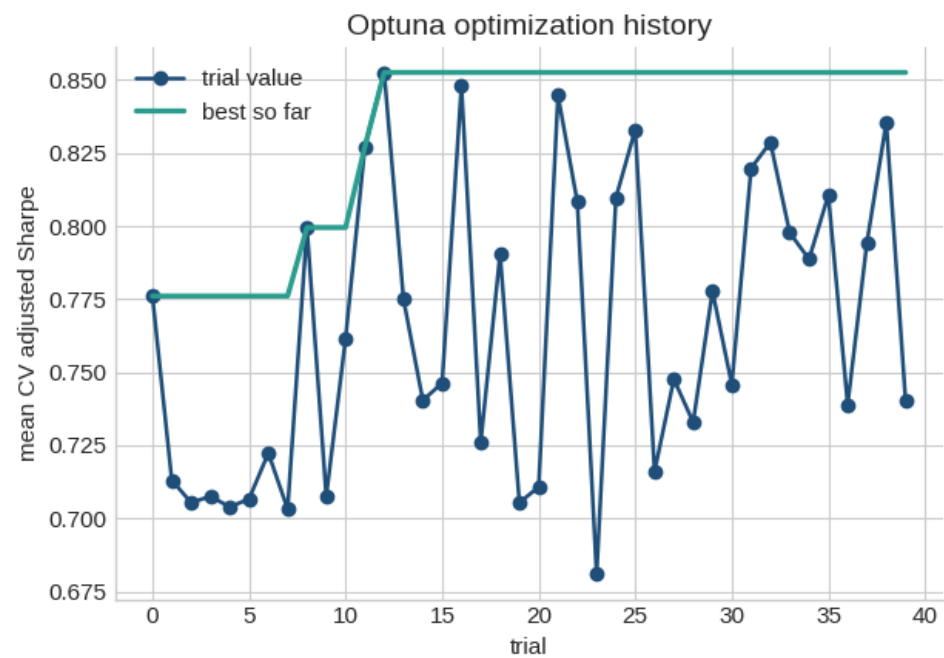
```
confidence_threshold    0.363733
gain                   179.320325
target_annual_vol      0.298329
soft_vol_cap           1.149479
min_weight             0.000000
max_weight             2.000000
vol_scale_min          0.500000
vol_scale_max          2.000000
vol_servo_window       60.000000
hard_vol_cap           1.200000
```

dtype: float64

Best inner CV mean adjusted Sharpe: 0.8526

Top trials

	number	mean_cv_adjusted_sharpe	params_confidence_threshold	params_gain	params_soft_vol_cap	params_target_annual_vol	status
12	12	0.852582	0.363733	179.320325	1.149479	0.298329	COMPLETED
16	16	0.848065	0.360912	161.752793	1.199203	0.298248	COMPLETED
21	21	0.844707	0.359024	161.139753	1.136887	0.299400	COMPLETED
38	38	0.835483	0.373504	162.926585	1.143599	0.270771	COMPLETED
25	25	0.833047	0.350165	136.314360	1.181092	0.277757	COMPLETED
32	32	0.828679	0.417709	167.597209	1.185818	0.278671	COMPLETED
11	11	0.827194	0.362772	165.226367	1.135585	0.291277	COMPLETED
31	31	0.819814	0.359759	134.347669	1.182927	0.278375	COMPLETED
35	35	0.810606	0.353334	128.539642	1.173502	0.241984	COMPLETED
24	24	0.809756	0.451731	178.284084	1.149499	0.248024	COMPLETED



```

In [29]: # 9. Refit the fixed modeling pipeline on the full tuning block, evaluate once on the untouched holdout, and build performance metrics.
# 9.1 Fit the full fixed modeling pipeline on all tuning rows.
feature_state = fit_feature_state(tune_df)
X_tune = transform_features(tune_df, feature_state, use_history=False)
X_holdout = transform_features(holdout_df, feature_state, use_history=True)

regime_state = fit_regime_detector(X_tune, returns=tune_df["forward_returns"], n_states=3)
reg_tune = predict_regimes(X_tune, regime_state)
reg_holdout = predict_regimes(X_holdout, regime_state)

X_tune_all = pd.concat([X_tune, reg_tune.drop(columns=["regime_label"])], axis=1)
X_holdout_all = pd.concat([X_holdout, reg_holdout.drop(columns=["regime_label"])], axis=1)

model_state = fit_return_model(X_tune_all, tune_df["market_forward_excess_returns"])
pred_holdout = predict_return_model(X_holdout_all, model_state)

# 9.2 Apply the best CV-selected strategy parameters to the untouched holdout.
holdout_weights = generate_weights(
    predicted_excess_return=pred_holdout["predicted_excess_return"],
    confidence=pred_holdout["confidence"],
    strategy_params=best_strategy_params,
    current_volatility=X_holdout.get("roll_vol_20", pd.Series(np.nan, index=X_holdout.index)),
    regime_label=reg_holdout["regime_label"],
    forward_returns=holdout_df["forward_returns"],
    risk_free_rate=holdout_df["risk_free_rate"],
)
baseline_weights = pd.Series(1.0, index=holdout_df.index)

holdout_solution = holdout_df[["risk_free_rate", "forward_returns"]].copy()
final_holdout_adjusted_sharpe = score(
    holdout_solution,
    pd.DataFrame({"prediction": holdout_weights.values}, index=holdout_df.index),
)
final_holdout_adjusted_sharpe_baseline = score(
    holdout_solution,
    pd.DataFrame({"prediction": baseline_weights.values}, index=holdout_df.index),
)

holdout_results = holdout_df[["date_id", "forward_returns", "risk_free_rate"]].copy()
holdout_results["prediction"] = holdout_weights.values
holdout_results["confidence"] = pred_holdout["confidence"].values
holdout_results["predicted_excess_return"] = pred_holdout["predicted_excess_return"].values

```

```
holdout_results["regime_label"] = reg_holdout["regime_label"].values
holdout_results["strategy_returns"] = (
    holdout_results["risk_free_rate"] * (1 - holdout_results["prediction"]) + holdout_results["prediction"] * holdout_re
)
holdout_results["baseline_returns"] = holdout_results["forward_returns"]
holdout_results["strategy_cumulative"] = (1 + holdout_results["strategy_returns"]).cumprod()
holdout_results["baseline_cumulative"] = (1 + holdout_results["baseline_returns"]).cumprod()
```

```

In [30]: # 10. Report the selected strategy parameters, the best inner CV score, the final holdout score, and the holdout plots.
print("Selected strategy parameters from Optuna")
display(pd.Series(best_strategy_params))
print(f"\nInitial untuned strategy mean CV adjusted Sharpe: {baseline_inner_cv_mean:.4f}")
print(f"Best inner CV mean adjusted Sharpe: {best_inner_cv_mean_adjusted_sharpe:.4f}")
print(f"Final holdout adjusted Sharpe (model): {final_holdout_adjusted_sharpe:.4f}")
print(f"Final holdout adjusted Sharpe (baseline w=1): {final_holdout_adjusted_sharpe_baseline:.4f}")
print("\nBest-CV fold diagnostics")
display(best_cv_results)
print("\nHoldout predictions preview")
display(holdout_results[["date_id", "prediction", "confidence", "predicted_excess_return", "regime_label"]].head(10))

fig, axes = plt.subplots(2, 2, figsize=(13, 9))
axes[0, 0].plot(holdout_results["date_id"], holdout_results["prediction"], color=COLORS["blue"], linewidth=2)
axes[0, 0].axhline(1.0, color=COLORS["gray"], linestyle="--", linewidth=1.2)
axes[0, 0].set_title("Holdout portfolio weight")
axes[0, 0].set_xlabel("date_id")
axes[0, 0].set_ylabel("weight")

axes[0, 1].plot(holdout_results["date_id"], holdout_results["confidence"], color=COLORS["gold"], linewidth=2)
axes[0, 1].axhline(best_strategy_params["confidence_threshold"], color=COLORS["red"], linestyle="--", linewidth=1.2, label="threshold")
axes[0, 1].set_title("Holdout prediction confidence")
axes[0, 1].set_xlabel("date_id")
axes[0, 1].set_ylabel("confidence")
axes[0, 1].legend()

axes[1, 0].plot(holdout_results["date_id"], holdout_results["strategy_cumulative"], color=COLORS["teal"], linewidth=2.4)
axes[1, 0].plot(holdout_results["date_id"], holdout_results["baseline_cumulative"], color=COLORS["gray"], linewidth=2.0)
axes[1, 0].set_title("Final holdout cumulative growth")
axes[1, 0].set_xlabel("date_id")
axes[1, 0].set_ylabel("growth of 1")
axes[1, 0].legend()

rolling_window = 60
rolling_strategy = holdout_results["strategy_returns"].rolling(rolling_window).mean() * 252
rolling_baseline = holdout_results["baseline_returns"].rolling(rolling_window).mean() * 252
axes[1, 1].plot(holdout_results["date_id"], rolling_strategy, color=COLORS["teal"], linewidth=2.2, label="strategy")
axes[1, 1].plot(holdout_results["date_id"], rolling_baseline, color=COLORS["gray"], linewidth=1.8, label="baseline w=1")
axes[1, 1].axhline(0.0, color=COLORS["red"], linestyle=":", linewidth=1)
axes[1, 1].set_title(f"Rolling {rolling_window}-day annualized return")
axes[1, 1].set_xlabel("date_id")
axes[1, 1].set_ylabel("annualized return")

```

```

axes[1, 1].yaxis.set_major_formatter(PercentFormatter(1.0))
axes[1, 1].legend()
plt.tight_layout()
plt.show()

```

Selected strategy parameters from Optuna

```

confidence_threshold      0.363733
gain                      179.320325
target_annual_vol        0.298329
soft_vol_cap              1.149479
min_weight                0.000000
max_weight                2.000000
vol_scale_min             0.500000
vol_scale_max             2.000000
vol_servo_window         60.000000
hard_vol_cap              1.200000

```

dtype: float64

Initial untuned strategy mean CV adjusted Sharpe: 0.7246

Best inner CV mean adjusted Sharpe: 0.8526

Final holdout adjusted Sharpe (model): 0.8605

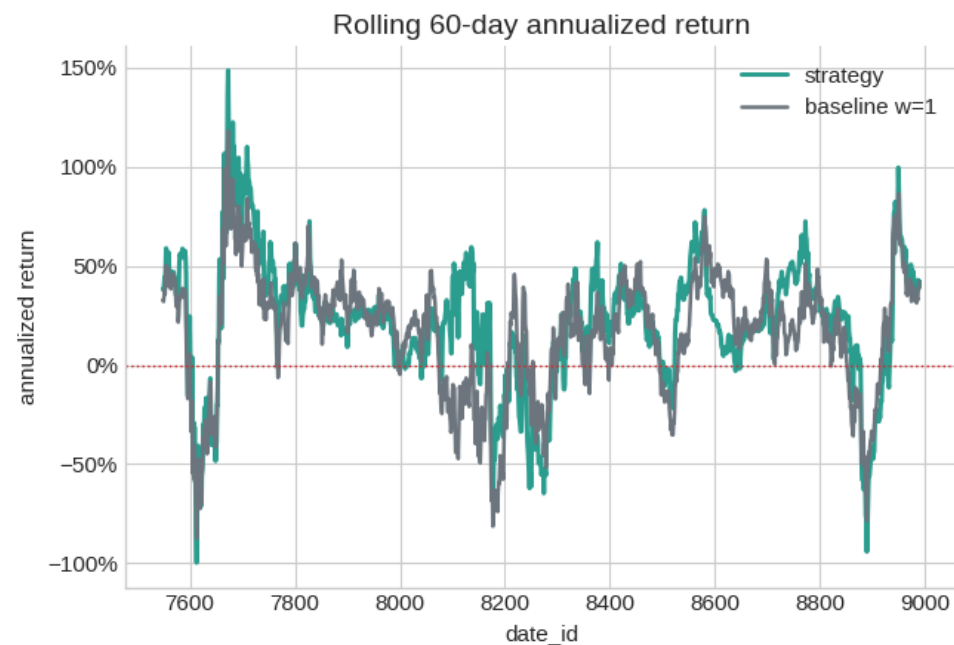
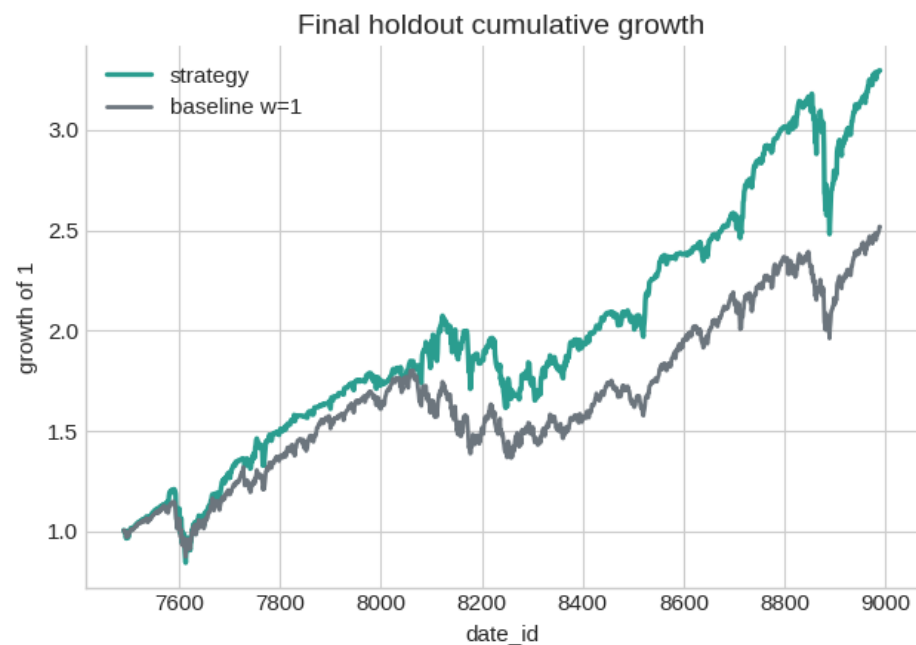
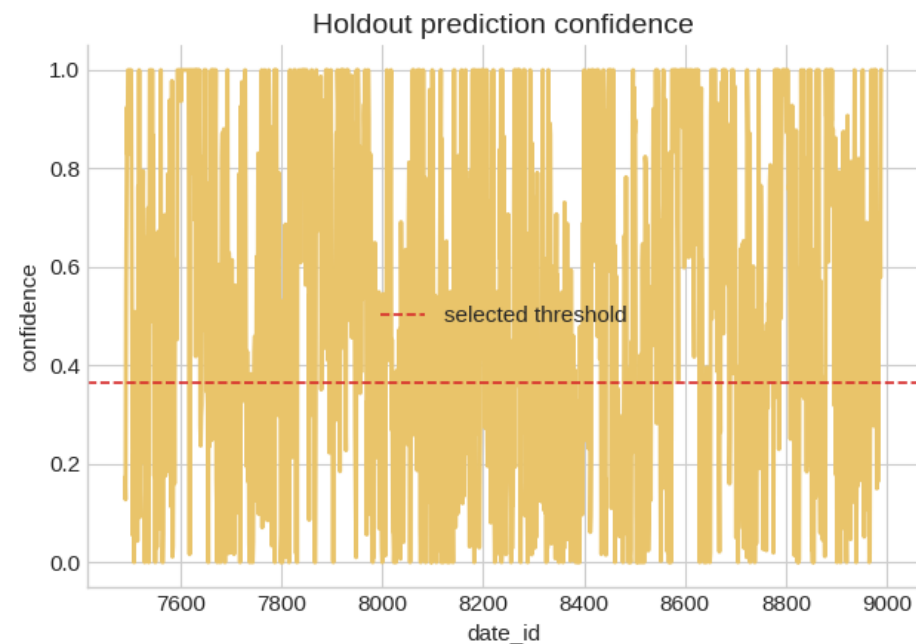
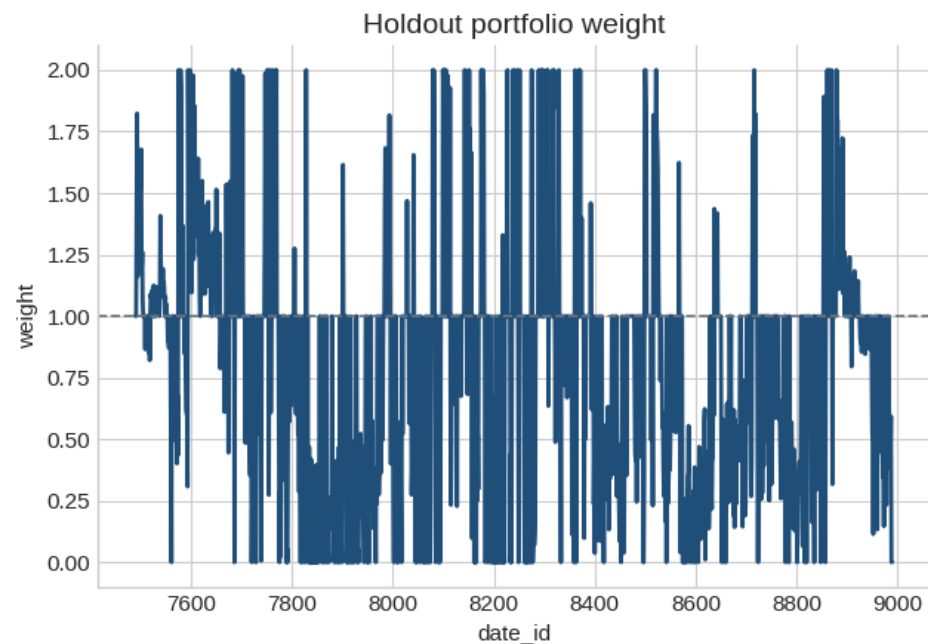
Final holdout adjusted Sharpe (baseline w=1): 0.7151

Best-CV fold diagnostics

	fold	train_start	train_end	val_start	val_end	adjusted_sharpe	mean_confidence	mean_weight	mean_roll_vol_20	dominant_regime
0	1	0	3644	3790	4372	0.998524	0.350858	0.913950	0.097941	transition
1	2	0	4227	4373	4955	-0.274084	0.423197	1.135643	0.237727	transition
2	3	0	4810	4956	5538	0.857809	0.369652	0.980680	0.177386	bear_high_vol
3	4	0	5393	5539	6121	1.570676	0.322459	1.012414	0.113536	transition
4	5	0	5976	6122	6704	0.966058	0.533008	0.949712	0.126502	bear_high_vol
5	6	0	6559	6705	7287	0.996510	0.887947	0.930480	0.097888	transition

Holdout predictions preview

	date_id	prediction	confidence	predicted_excess_return	regime_label
7490	7490	1.000000	0.172498	0.001022	transition
7491	7491	1.000000	0.127569	0.000945	bull_low_vol
7492	7492	1.459473	0.540532	0.003621	bear_high_vol
7493	7493	1.823903	0.739317	0.004505	bull_low_vol
7494	7494	1.161733	0.924707	0.006040	bear_high_vol
7495	7495	1.386221	0.828207	0.005664	bull_low_vol
7496	7496	1.483112	1.000000	0.019123	bear_high_vol
7497	7497	1.522730	1.000000	0.007901	bull_low_vol
7498	7498	1.187570	1.000000	0.007591	bear_high_vol
7499	7499	1.550310	1.000000	0.008252	bull_low_vol



Interpretation

The Optuna search used only the inner walk-forward CV folds built from the tuning block. The final holdout score above is evaluated once after parameter selection and is the notebook's unbiased offline estimate. `data/test.csv` is not used anywhere in this workflow.